Controller Design

High-level design

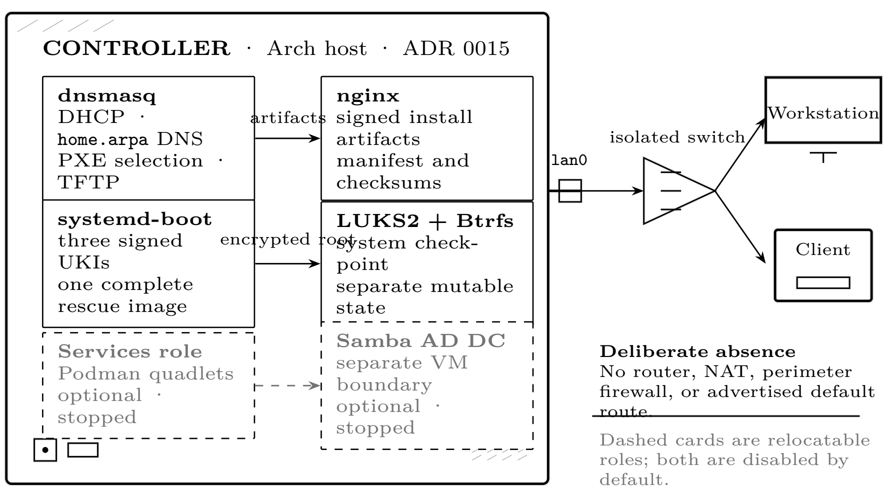
What the Controller is, what it owns, and what it deliberately does not

The Controller is the homelab's infrastructure host. It answers DHCP and DNS for its managed network, serves the boot chain that installs every other machine, and is the one system that must be reconstructible without any other system being available. Everything else in the lab may depend on the Controller. The Controller may depend on nothing except a copy of this document, an installer image and a recorded set of inputs.

Scope

It owns	Boot, storage and encryption; the managed network interface; bundled DHCPv4 and `home.arpa` DNS; PXE selection and first-stage TFTP; the HTTP artifact service; local recovery; and its own checkpoint and rollback.
It may host	The optional Services role — Homebridge, openHAB and similar — and the identity domain controller in a VM. Both are separable and neither is required for the Controller to do its job. GOVERNED BY ADR 0054, ADR 0055
It does not do	Routing, NAT or perimeter firewalling. It has one network interface and advertises no default route. On the isolated acceptance network there is deliberately no path to the Internet. GOVERNED BY ADR 0011
It is not	A hypervisor-first platform. Foundational services run directly on the Arch host; containers and VMs are used where they earn their place, not by default. GOVERNED BY ADR 0014
One failure domain	Containers and VMs on the Controller share its hardware, its kernel maintenance window and its power supply. Placing a service on the Controller does not make it independently available.

Roles and boundaries



Storage

Part	Size	Why
ESP	2 GiB, FAT32	systemd-boot plus three UKIs, one of which carries a complete rescue userspace. Sized with deliberate headroom so an update can stage a full new artifact set before switching. The ESP is the one partition that cannot be grown afterwards without moving everything behind it. GOVERNED BY ADR 0051
Root	remainder, LUKS2	Btrfs inside the encrypted container. The operating system is one checkpointable subvolume; logs, caches, home directories, the snapshot store and mutable service data are separate subvolumes excluded from it. GOVERNED BY ADR 0026, ADR 0027
Swap	file in root	A file inside the encrypted Btrfs, so encrypted swap costs no second LUKS container. Hibernation is disabled — a machine whose job is answering DHCP does not suspend.

There is no Windows partition. ADR 0047 removed it: firmware maintenance uses `fwupd` where the vendor publishes to LVFS and vendor-supplied bootable media otherwise, which costs nothing until the day it is needed and removes a licence, a BitLocker recovery-key custody chain and a second Secure Boot trust path from the design.

The network boundary

One interface	Selected explicitly at install time and pinned by permanent MAC address to the stable name `lan0`. There is no default and no automatic choice, because binding DHCP to the wrong link puts a second DHCP authority on a network the Controller does not own. GOVERNED BY ADR 0050
One authority	dnsmasq is the only address-assigning DHCP server on its layer-2 network. Where external infrastructure owns DHCP, the Controller's instance stays stopped. Two authorities on one segment is forbidden at every stage, including transitions. GOVERNED BY ADR 0008, ADR 0044
Bundled	DHCP and DNS start and stop as one unit. A partial success is a failure. GOVERNED BY ADR 0012
IPv4 only	Managed addressing is IPv4. The host's IPv6 stack stays enabled and unmanaged; nothing globally disables it. GOVERNED BY ADR 0013
Four inputs	The operator supplies subnet, Controller address, and pool start and end. Netmask, network and broadcast addresses, the DNS server, the suffix and the absence of a router are all derived. Contradictory values cannot reach dnsmasq because they cannot be entered. GOVERNED BY ADR 0045
Fail closed	First boot verifies the interface, its MAC, carrier, the address and the absence of a competing DHCP authority before starting anything. If any check fails, DHCP stays stopped and the machine reports why. GOVERNED BY ADR 0009

Lifecycle: install once, converge continuously

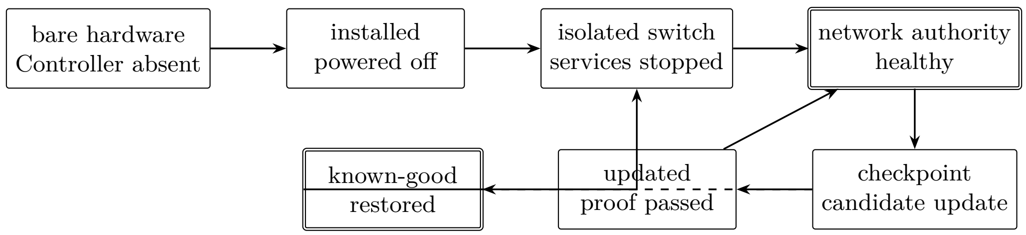
The single most important structural decision is that installation and configuration are different things with different cadences.

Stage	What happens
Install	Only what cannot be done later: partition, encrypt, make filesystems, install the base system and both kernels, write boot artifacts, configure `lan0` and the static address, install `sshd` with the administrator public key, record the manifest. Then power off. GOVERNED BY ADR 0010, ADR 0053
Relocate	The operator physically moves the powered-off Controller to the isolated switch. Being powered off during the move is the primary DHCP-conflict safety boundary — not a configuration flag. GOVERNED BY ADR 0010
First boot	One-shot activation validates the network plan, then starts dnsmasq and nginx together, or leaves them stopped and says why.
Converge	Everything else is Ansible over SSH from this repository: package sets, service configuration, the Services role, identity client, users, sudo, health checks. Idempotent, reviewable as a diff, and the same mechanism for the Controller and every Workstation. GOVERNED BY ADR 0053
Update	Guarded: read the news, checkpoint, full-system upgrade, reboot if foundational, validate DHCP, DNS and PXE automatically, roll back if validation fails. GOVERNED BY ADR 0016, ADR 0028

Why this split matters

The previous design specified provisioning in detail and day-two configuration not at all, which meant that adding a service or changing a DHCP reservation implied reinstalling the machine. Nothing iterates at that cost, so in practice the running system drifts from the record and the record stops being true. The install stage is deliberately small so that it can be fully tested; everything that changes often lives where changes are cheap and reversible.

State model: prove each boundary before crossing it



Every arrow has an evidence requirement. "Command returned zero" is never enough: the observation must establish the property named on the arrow.

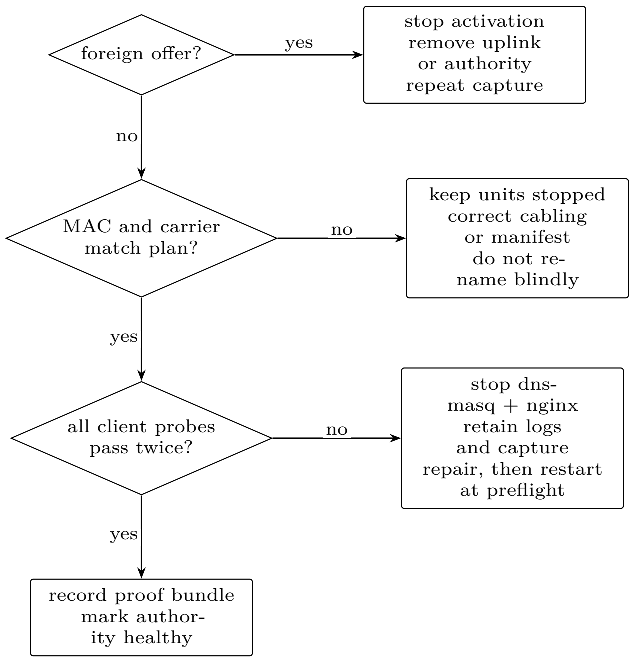
Transition	Operator question	Required evidence
Absent → installed	Is this the intended chassis, disk serial and permanent MAC?	Printed and on-disk manifests agree; the selected disk is no longer offered as ambiguous; shutdown, not reboot, is the final event.
Installed → placed	Is the destination segment physically isolated from every other DHCP authority?	Switch-port inventory or cable trace; Controller remains powered off until attached; no upstream/router port is present.
Placed → authority	Do carrier, MAC, address and DHCP-silence checks all describe this exact segment?	Preflight transcript; packet capture covering the listen window; bound sockets; one test lease and matching forward/reverse DNS.
Authority → updated	Can clients still boot while stateful roles remain intact?	Checkpoint identifier; package transaction; cold-boot record; DHCP/DNS/PXE probes; service-state hashes and application-specific health.
Failure → restored	Did rollback restore the previous system without rewinding mutable data?	Booted checkpoint identity; excluded-subvolume inventory; repeated network probes; incident note retaining the failed candidate.

Activation runbook

1. With power still off, trace `lan0` to the isolated switch. Ask: *What observation would reveal an accidental uplink?* Record the answer and the port identifiers before applying power.
2. Boot locally and unlock LUKS. Record the manifest mode, root subvolume, kernel command line and permanent MAC. A mismatch stops activation.
3. Keep dnsmasq and nginx stopped. Observe the segment long enough to see ordinary client discovery traffic; any DHCPOFFER from another source is a hard stop. Silence is evidence only when the capture interface and filter are also recorded.
4. Run the first-boot preflight. Compare its derived network, broadcast, pool and Controller address with the printed plan. The operator explicitly answers: *Is the Controller address outside the pool? Is no router option present?*
5. Start the bundled network unit. From a sacrificial client, renew a lease, resolve the Controller name and reverse address, fetch the manifest and begin one PXE boot. Save server logs and the client's observed values.
6. Reboot the Controller from cold power. Repeat the same probes without editing configuration. Only this second pass establishes persistence.

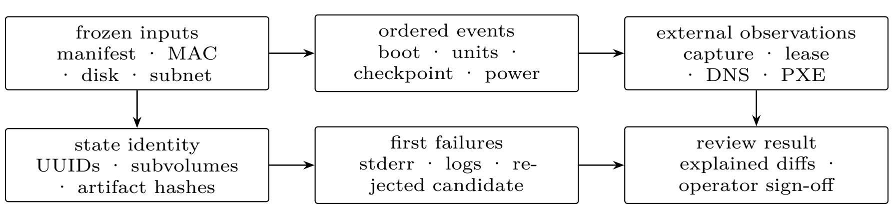
Check	Command or observation	Pass evidence
Interface identity	`networkctl status lan0`; permanent address from `ip -d link`	Configured MAC equals observed permanent MAC; exactly one managed link.
No competing DHCP	packet capture of UDP 67/68 before activation	No DHCPOFFER except the Controller after activation; capture duration and interface recorded.
Bound services	socket and unit inventory	dnsmasq bound only to `lan0`; nginx serves only the intended artifact root; both units active together.
Lease and names	client renew; forward and reverse queries	Address inside pool, no default gateway, DNS points to Controller, forward and reverse answers agree.
Boot service	client PXE request and artifact fetch	TFTP transfers only iPXE; HTTP checksum equals manifest; requested artifact is the current named build.
Recovery	boot rescue UKI with normal root unavailable	Rescue userspace reaches local console and can inspect/unlock the encrypted root without network service.

Failure branches



Never "test around" a failed bundle member by leaving the other running. A partial DHCP/DNS/PXE service is a failed Controller, and its evidence bundle must retain the first failure rather than only the successful retry.

Proof-bundle anatomy



The bundle is complete only when it connects declared inputs to observations made outside the Controller. Server logs corroborate client evidence; they do not replace it. Secrets, passphrases and private key material never enter the bundle.

Optional roles

Services	Homebridge, openHAB and similar, as Podman quadlets with host networking — required, because HomeKit and UPnP discovery depend on mDNS and do not survive bridged container networking. USB radios bound by stable `by-id` path. Application state on its own subvolume with its own backup, because its restore point is not the operating system's. Disabled by default. GOVERNED BY ADR 0054
The trade-off	Running Services on the Controller means a Controller maintenance window is also a home-automation outage. That is accepted and recorded, not hidden. When it stops being acceptable, the role moves to its own host and nothing else changes.
Identity	Samba AD DC in a VM — a genuine separate trust boundary with its own restart and upgrade cadence, which is precisely when ADR 0014 permits a VM. Arch clients use SSSD's AD provider; Windows Pro clients domain-join; an AD-capable NAS can share the same identities. GOVERNED BY ADR 0055
Break-glass	Every managed machine keeps a separately named local administrator that is never a directory account. While the directory is down, only cached and break-glass logins work — so this is mandatory, not a nicety.

Milestones

A: functional proof	Network boot, interactive authorization, install, isolated-network activation, DHCP and DNS validation, checkpoint and rollback, Workstation pivot. Encryption is real throughout. Secure Boot may be disabled under an explicit, recorded `development-proof` mode with manual LUKS2 unlock at every boot, no production data, and no custom signing material anywhere. GOVERNED BY ADR 0043
B: hardening	Revisit ADRs 0029–0042 as a group with operational experience, then enforce Secure Boot, implement signing and custody, enable TPM unlock only after its recovery path is tested, and prove an unattended cold restart restores service. Reprovision from the proven installer; the proof installation is disposable by construction. GOVERNED BY ADR 0043
Known defects to fix in B	The key separation in ADRs 0033–0035 is notional once ADR 0039 gives both media one passphrase; the backup's independence is defeated by sharing that passphrase; and ADR 0030's additive trust model may be infeasible on firmware that only permits `db` changes from Setup Mode. All three are recorded rather than rediscovered. GOVERNED BY ADR 0057
No system passes	Final acceptance until Milestone B is complete.

Measurable acceptance

Property	Threshold	Final proof
Bootstrap independence	Zero Controller-hosted prerequisites used during a bare-metal rebuild	Start with Controller powered off and storage unavailable; rebuild from offline instructions, installer and recorded inputs.
Network authority	Zero foreign offers; exactly one Controller offer per successful discovery	Timestamped capture spanning preflight and three client renewals, plus lease and DNS records.
Cold recovery	Three consecutive cold boots return DHCP, DNS and PXE without intervention after LUKS unlock in Milestone A	Power-event log and identical automated probe reports. Milestone B additionally requires the approved unattended unlock path.
Rollback	Failed system candidate reverted within 30 minutes; mutable service state neither rewound nor silently advanced by the rollback	Before/after checkpoint IDs, subvolume generations, probe report and application-state check.
Rebuild fidelity	Rebuilt manifest matches recorded role, disk, MAC and network inputs; every declared package/service difference is explained	Manifest diff, configuration diff, acceptance transcript and signed-off exceptions.

Safety boundary	No production data or production signing material on a `development-proof` system	Storage inventory and manifest-mode assertion included in the proof bundle.

FINAL PROOF IS A BUNDLE, NOT A VERDICT

Acceptance consists of the input manifest, cable/port record, preflight and packet capture, checkpoint identifiers, cold-boot transcripts, client probe results, recovery exercise and explained diffs. It passes only when a second operator can trace every claimed boundary to an observation and reproduce the probe without privileged knowledge.

THE BOOTSTRAP RULE

The Controller rebuild procedure must never require a functioning Controller to obtain the instructions, DNS records, installer image, artifact manifest or decryption material needed to rebuild it. Every recovery path in this design is tested with the Controller assumed dead. That is the whole reason this document is printable.

Final review worksheet

Reviewer must answer	Evidence reference or disposition
Can the Controller be rebuilt while its disks, DNS and artifact service are unavailable?	Offline-media run ID, manifest digest and rebuild transcript:
Did any foreign DHCP authority answer during preflight or activation?	Capture file, interface and observation window:
Do client observations agree with server logs for lease, DNS and PXE?	Client probe report and correlated log interval:
Did three cold boots restore the same declared authority without configuration edits?	Power-event IDs and probe reports:
Did rollback preserve excluded mutable state and restore the named system checkpoint?	Before/after subvolume and checkpoint identities:
Are all differences, waivers and failed candidates retained rather than silently normalized?	Review record, owner and expiry:

Disposition: ☐ accept ☐ reject ☐ repeat named proof Reviewer / date: ____________